



Institut Universitaire des sciences

- Faculté des sciences et de technologie

✓ TD N°6 Réseau II

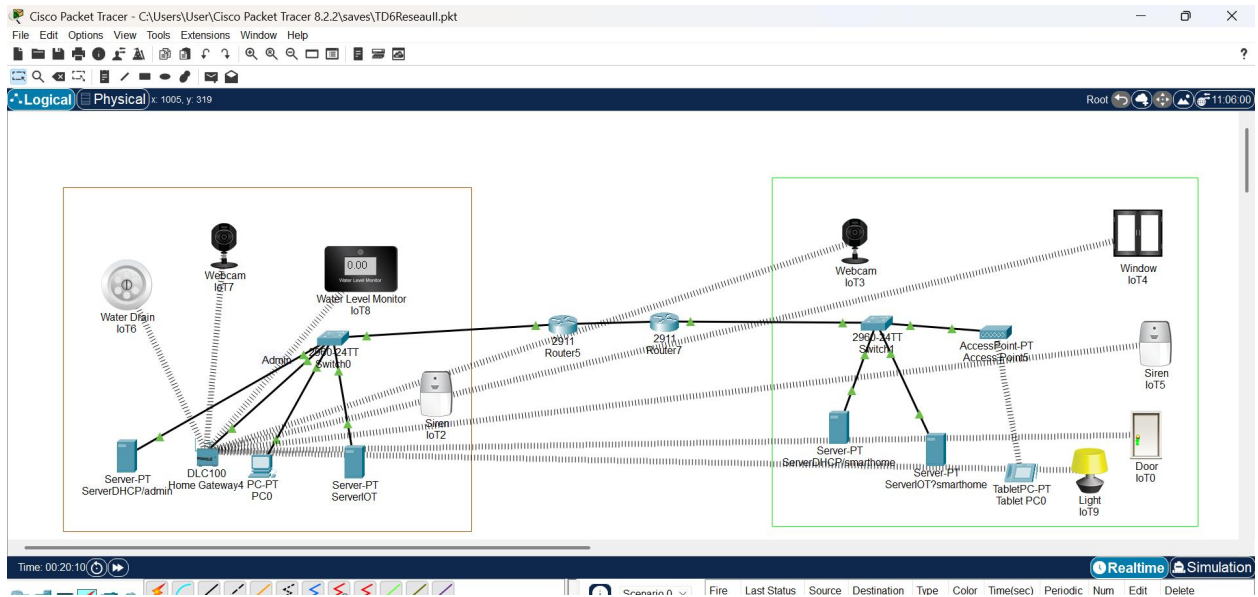
Nom & Prénom : COFFY Cliford

Niveau : L3

Prof. : Ismaël SAINT AMOUR

Date : 2/05/2026

Creation de la Topologie



Router5

Physical

Config

CLI

Attributes

MODULES

HWIC-1GE-SFP

HWIC-2T

HWIC-4ESW

HWIC-8A

WIC-Cover

GLC-LH-SMD

Physical Device View

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View



Customize Icon in Logical View



The HWIC-1GE-SFP is a single-wide HWIC with one Small Form-Factor Pluggable (SFP) slot. The SFP slot can be populated with Cisco copper and optical Gigabit Ethernet SFPs to provide 1-port Gigabit Ethernet connectivity on all Cisco Integrated Services Routers



☐ Top



Router5



Physical

Config

CLI

Attributes

MODULES

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GLC-LH-SMD

Physical Device View

Zoom In

Original Size

Zoom Out



Customize
Icon in
Physical View



Customize
Icon in
Logical View



The HWIC-2T is a Cisco 2-Port Serial High-Speed WAN Interface Card, providing 2 serial ports.



☐ Top

Router 1

 Router5

Physical Config CLI Attributes

IOS Command Line Interface

249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/0/0	unassigned	YES	unset	administratively down	down
Serial0/0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
%Invalid interface type and number
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#

Copy Paste

☐ Top

Routeur entre les deux routeurs (OSPF)

 Router5

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

```
Router>show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       unassigned      YES unset  administratively down  down
GigabitEthernet0/1       unassigned      YES unset  administratively down  down
GigabitEthernet0/2       unassigned      YES unset  administratively down  down
Serial0/0/0              unassigned      YES unset  administratively down  down
Serial0/0/1              unassigned      YES unset  administratively down  down
Vlan1                    unassigned      YES unset  administratively down  down
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
%Invalid interface type and number
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
Router(config-router)#
```

Copy

Paste

☐ Top

Router 2

 Router7

Physical

Config

CLI

Attributes

IOS Command Line Interface

2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

System Configuration Dialog

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/0/0	unassigned	YES	unset	administratively down	down
Serial0/0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface Serial0/0/0

Router(config-if)#ip address 192.168.1.2 255.255.255.252

Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down

Router(config-if)#exit

Router(config)#interface GigabitEthernet0/0

Router(config-if)#ip address 192.168.3.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#

Copy

Paste

☐ Top

Routeur entre les deux routeurs (OSPF)

 Router7

Physical Config CLI Attributes

IOS Command Line Interface

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>show ip interface brief
Interface                IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0       unassigned      YES unset  administratively down down
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Serial0/0/0              unassigned      YES unset  administratively down down
Serial0/0/1              unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/0/0
Router(config-if)#ip address 192.168.1.2 255.255.255.252
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

Copy Paste

☐ Top

Configuration des switchs

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#name Admin
      ^
% Invalid input detected at '^' marker.

Switch(config)#interface f0/1
Switch(config-if)#name Admin
      ^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#vlan 10
Switch(config-vlan)#name Admin
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name DMZ
Switch(config-vlan)#interface range fa0/1 - 10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#interface fa0/24
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
```

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☐ Top

Connection Type

Configuration des serveurs DHCP pour attribuer des IP automatiquement et le serveur IOT

The screenshot displays the Cisco Packet Tracer interface with the configuration window for a device named "Server-IOT?smarthome". The window is divided into several tabs: Physical, Config, Services, Desktop, Programming, and Attributes. The "Config" tab is active, showing the configuration for the "Server-IOT?smarthome" device.

The configuration is organized into three main sections:

- IP Configuration:**
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 192.168.1.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:**
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: (empty field) / (empty field)
 - Link Local Address: FE80::290:21FF:FEE0:6AA7
 - Default Gateway: (empty field)
 - DNS Server: (empty field)
- 802.1X:**
 - ☐ Use 802.1X Security
 - Authentication: MD5
 - Username: (empty field)
 - Password: (empty field)

The background shows a network diagram with a "Water Drain IoT6" device connected to a "Server-PT ServerDHCP/admin" device. The time displayed at the bottom is 01:43:26.

Cisco Packet Tracer

File Edit Options View Help

Logical

Physical

Water Droplet IoT6

Server DHCP

Time: 01:45:33

ServerDHCP/smarthome

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

DHCP

Static

IPv4 Address192.168.1.3

Subnet Mask255.255.255.0

Default Gateway192.168.1.1

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local AddressFE80::201:64FF:FEA8:79ED

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

2960-24TT Switch

smarthome

Server DHCP

Server IoT7smar

Last Status Source

Physical Config **Services** Desktop Programming Attributes**SERVICES**

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface Service ☒ On ☐ Off

Pool Name

Default Gateway

DNS Server

Start IP Address :

Subnet Mask:

Maximum Number of Users :

TFTP Server:

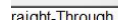
WLC Address:

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	0.0.0.0	192.168....	255.255....	251	0.0.0.0	0.0.0.0



Test sur les VLANs

Cisco Packet Tracer - C:\Users\User\Cisco Packet Tracer 8.2.2\saves\TD6Reseau.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x 382, y: 551

Root 00:18:00

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Water Drain IoT6

Webcam IoT7

Water Level Monitor IoT8

Adm IoT9

24TT Switch0

Server-PT verDHCPadmin

Home Gateway4

PC-PT PC0

Server-PT ServerIoT

Siren IoT2

Window IoT4

Siren IoT5

Door IoT0

Time: 00:43:50

Simulation

Edit Delete

Copper Straight-Through

Tracer - C:\Users\User\Cisco Packet Tracer 8.2.2\saves\TD6Reseau.pkt

File Edit Options View Tools Extensions Window Help

Physical x 647, y: 380

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Water Drain IoT6

Webcam IoT7

Water Level Monitor IoT8

Adm IoT9

24TT Switch0

Server-PT verDHCPadmin

Home Gateway4

PC-PT PC0

Server-PT ServerIoT

Siren IoT2

Window IoT4

Siren IoT5

Door IoT0

Time: 00:43:50

Simulation

Edit Delete

Copper Straight-Through

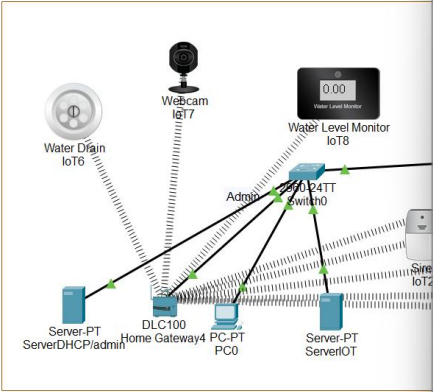
Top

Configuration du serveur IoT

Cisco Packet Tracer - C:\Users\User\Cisco Packet Tracer 8.2.2\saves\TD6ReseauII

Edit Options View Tools Extensions Window Help

Logical Physical x 1675, y 541



Water Drain IoT6
Webcam IoT7
Water Level Monitor IoT8
24TT Switch
Server-PT ServerDHCPadmin
Home Gateway4
PC-PT PC0
Server-PT ServerIoT

00:41:45

Copper Straight-Through

ioT9

Specifications Physical Config Attributes

GLOBAL
Settings
Algorithm Settings
Files
INTERFACE
Wireless0
Bluetooth

Wireless0

Port Status ☒ On

Bandwidth 300 Mbps

MAC Address 0007.ECCC.B2B1

SSID HomeGateway

Authentication
☒ Disabled
☐ WPA-PSK
☐ WPA
☐ 802.1X

☐ WEP
☐ WPA2-PSK
☐ WPA2
Method:

WEP Key
PSK Pass Phrase
User ID
Password
MD5
User Name
Link Local Address: FE80::207:ECFF:FECC:B2B1

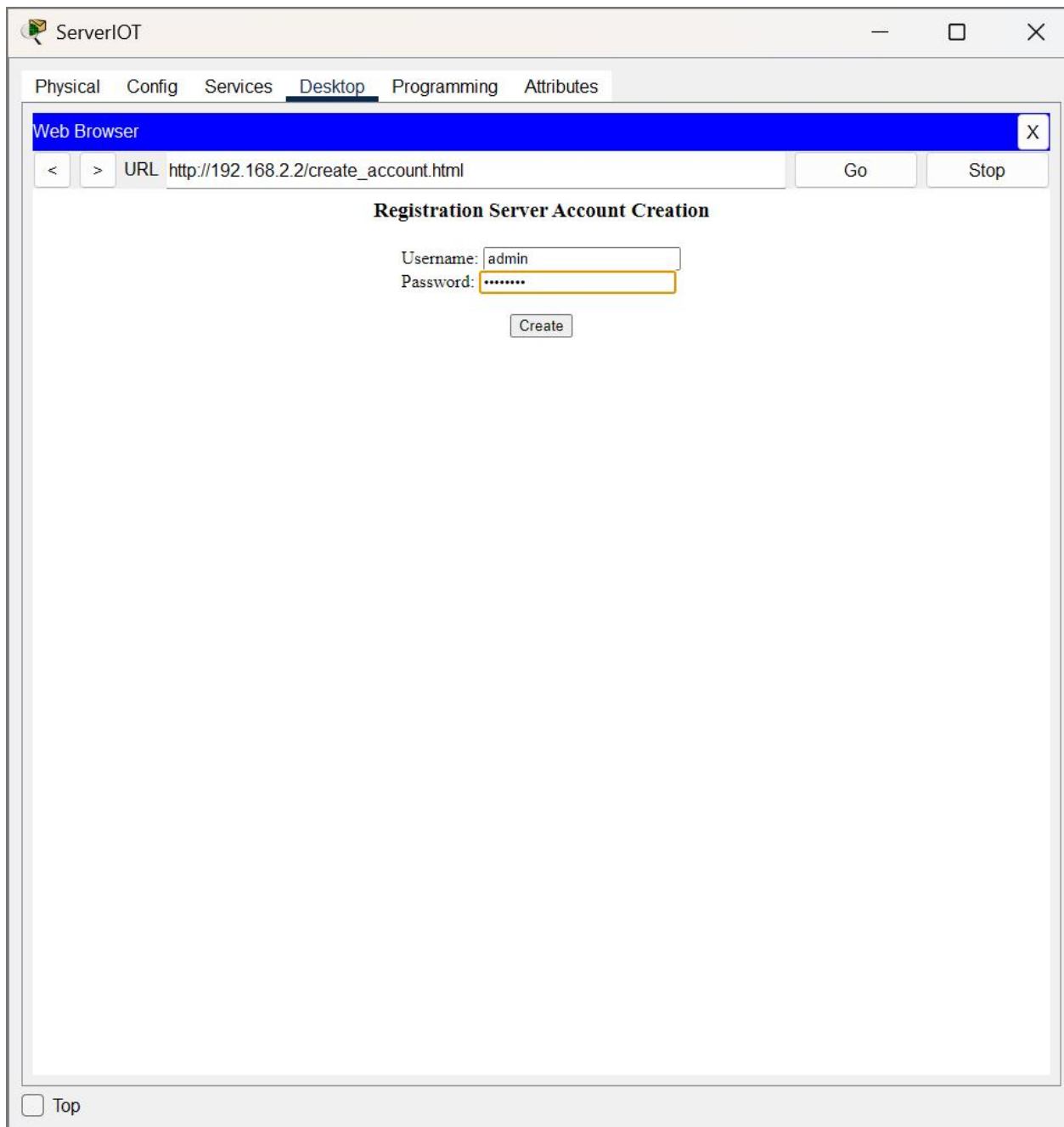
Encryption Type Disabled

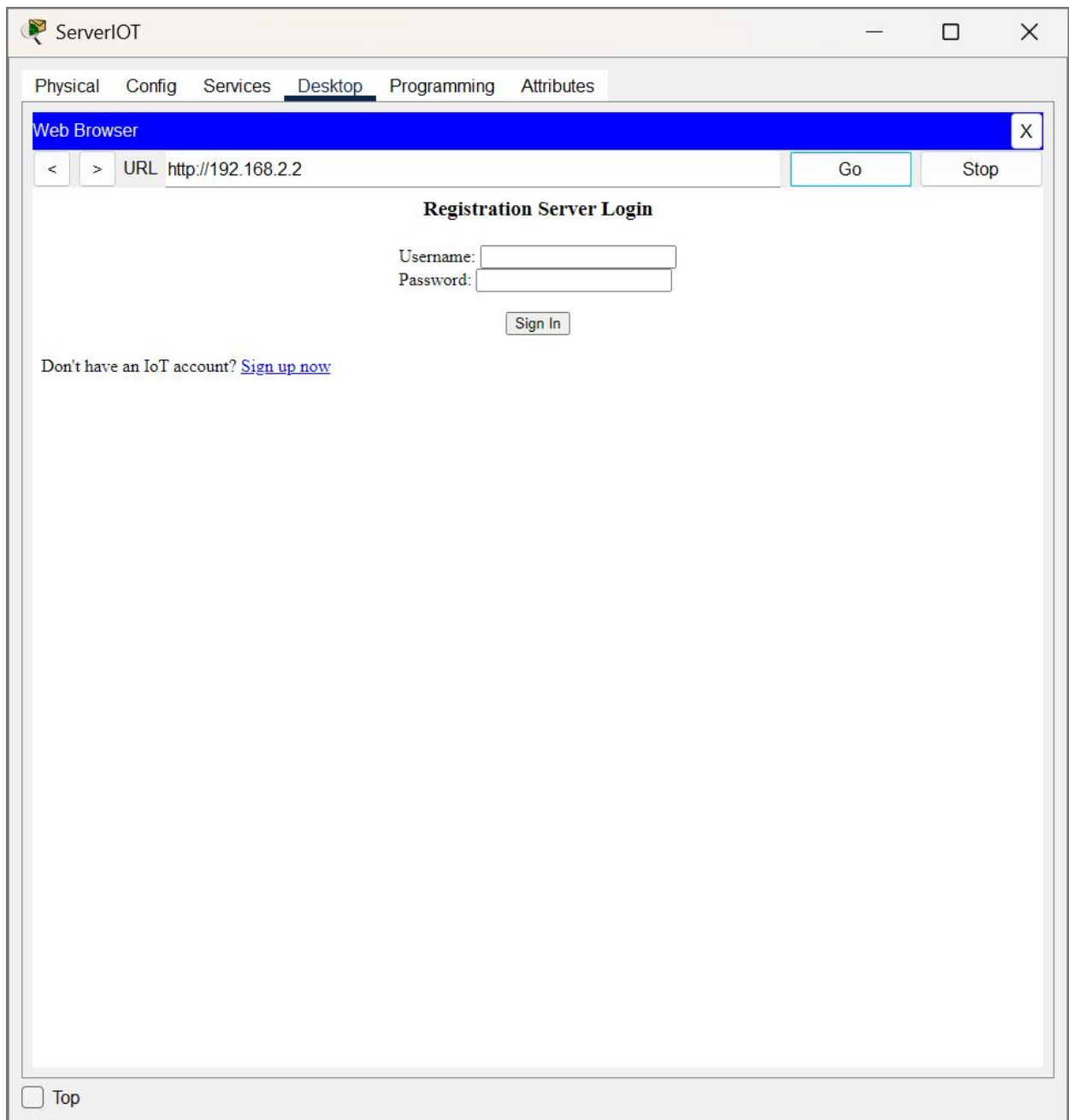
IP Configuration
☒ DHCP
☐ Static
IPv4 Address 192.168.25.100
Subnet Mask 255.255.255.0

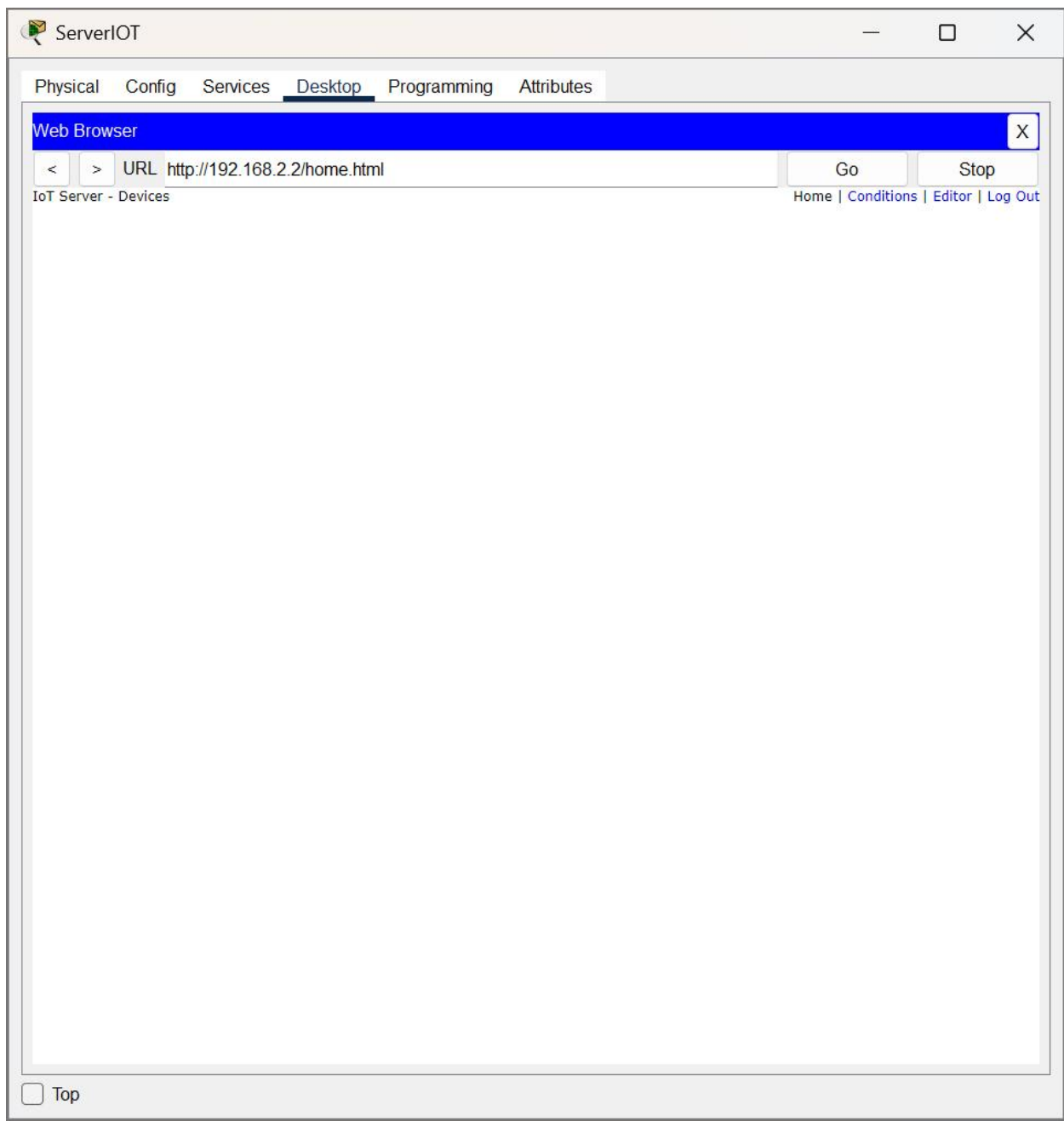
IPv6 Configuration
☒ Automatic
☐ Static
IPv6 Address
Link Local Address: FE80::207:ECFF:FECC:B2B1

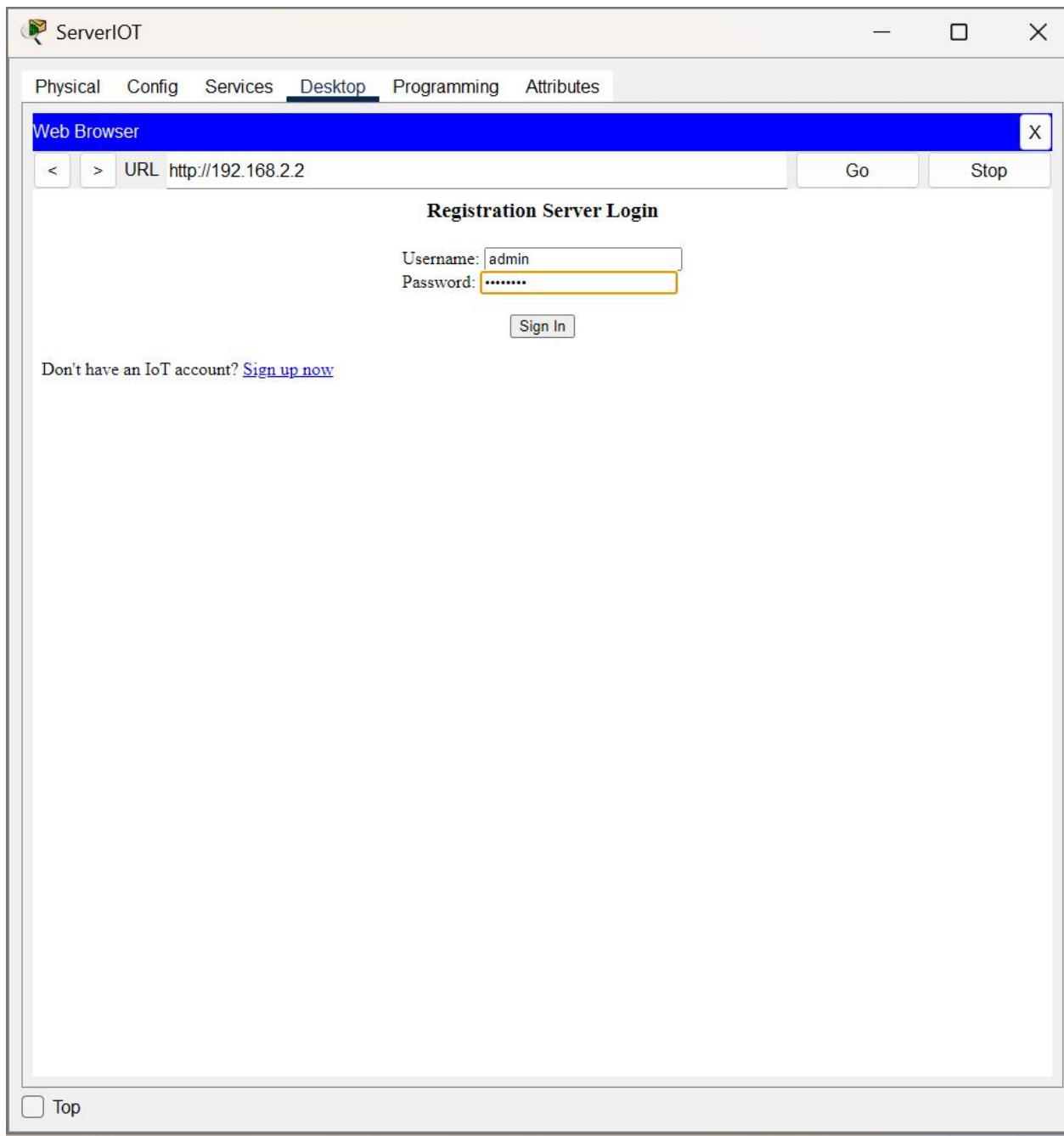
☐ Top

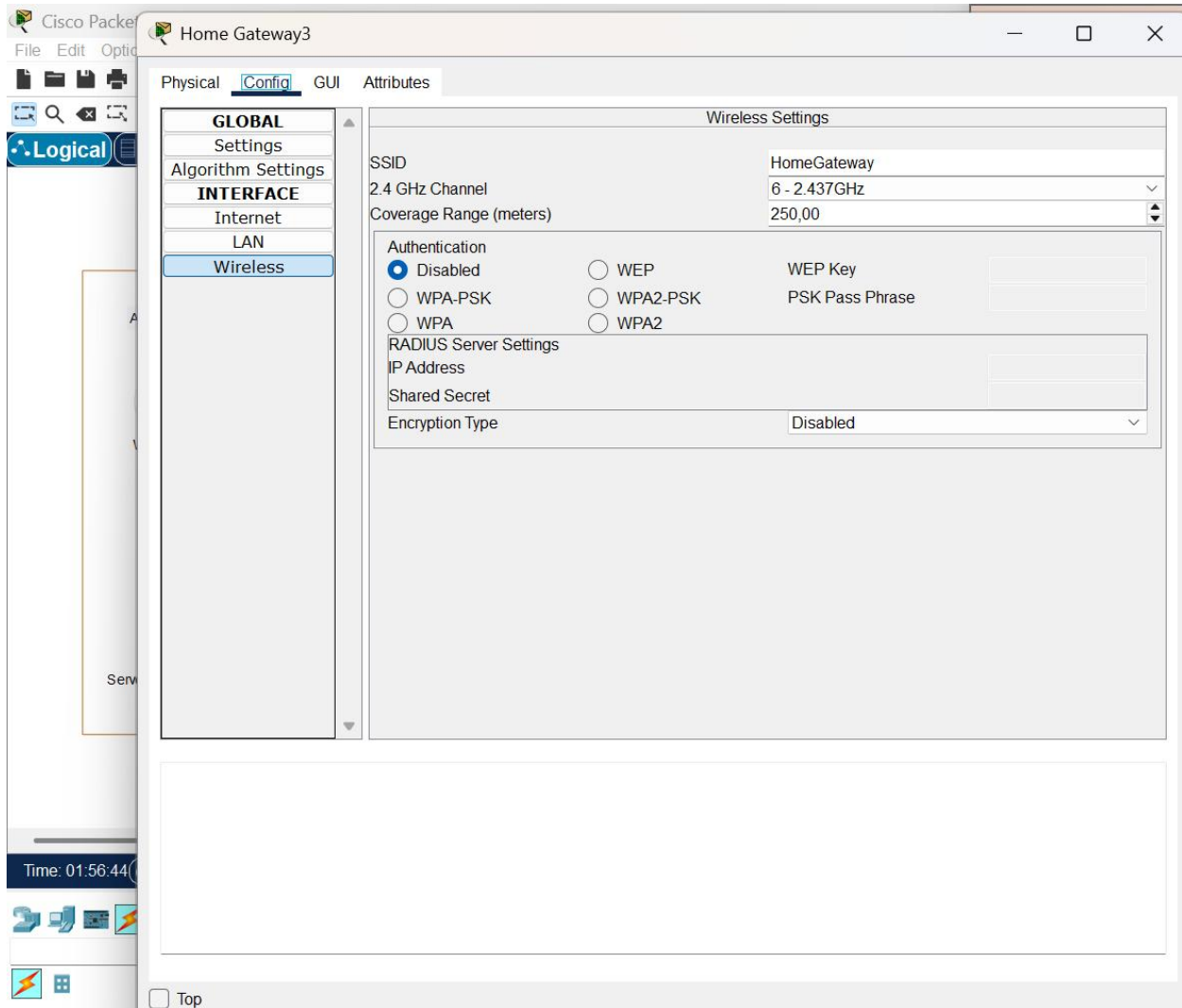
Registration au server



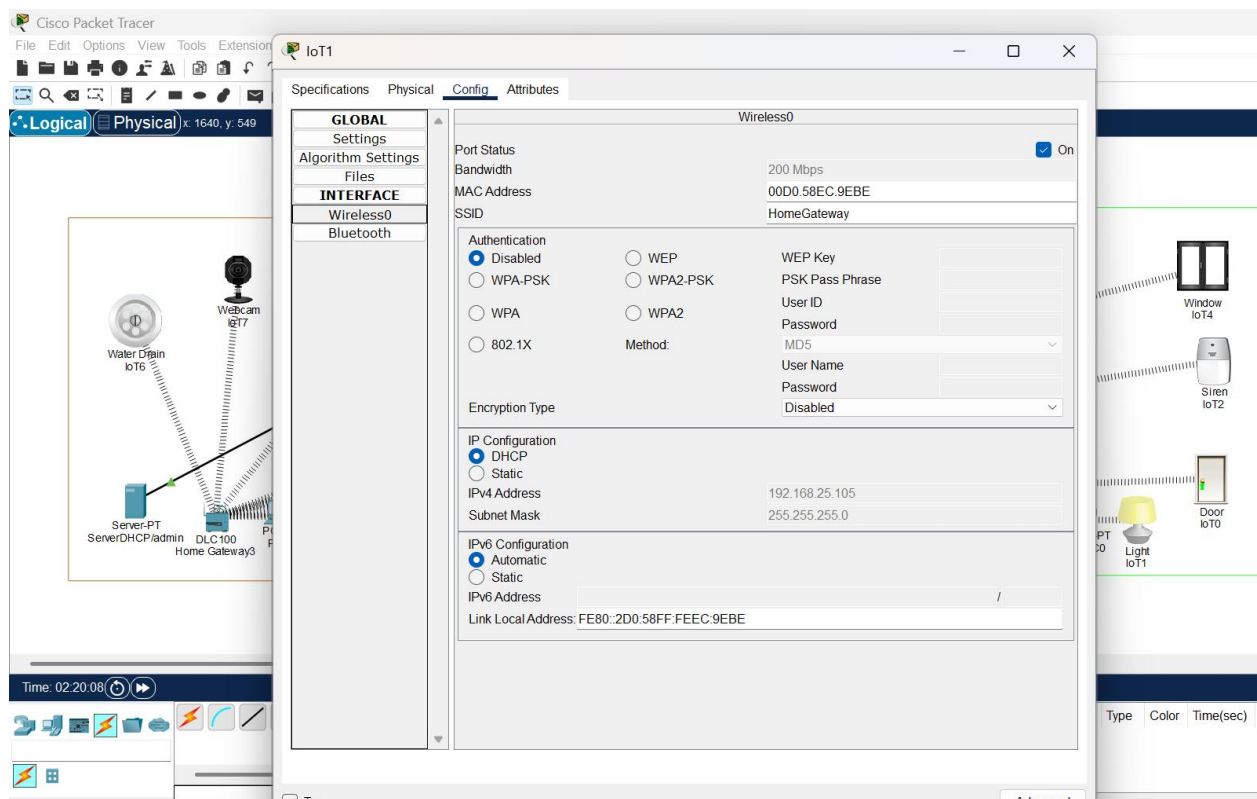








Configurer les IOT



IoT1

SpecificationsPhysicalConfigAttributes

GLOBAL

Settings

Algorithm Settings

Files

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status

200 Mbps

On

Bandwidth

200 Mbps

MAC Address

00D0.58EC.9EBE

SSID

lan1

Authentication

☒ Disabled

☐ WPA-PSK

☐ WPA

☐ 802.1X

☐ WEP

☐ WPA2-PSK

☐ WPA2

Method:

WEP Key

PSK Pass Phrase

User ID

Password

MD5

User Name

Password

Disabled

Encryption Type

Disabled

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

192.168.25.105

Subnet Mask

255.255.255.0

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

/

Link Local Address

FE80::2D0:58FF:FEEC:9EBE

☐ Top

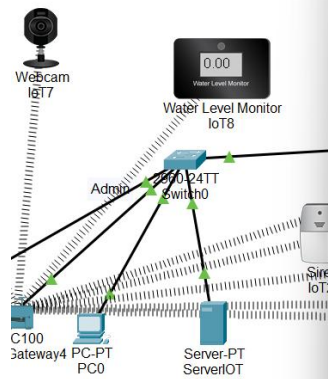
Advanced

r:\Cisco Packet Tracer 8.2.2\saves\TD6Reseau...

nsions Window Help



75



Copper Straight-Through

IoT0

SpecificationsPhysicalConfigAttributes

GLOBAL

Settings

Algorithm Settings

Files

INTERFACE

Wireless0

Bluetooth

Global Settings

Display Name Iot0

Serial Number PTT0810P9ZM-

Interfaces Wireless0

Gateway/DNS IPv4

DHCP

Static

Default Gateway 192.168.25.1

DNS Server 0.0.0.0

Gateway/DNS IPv6

Automatic

Static

Default Gateway

DNS Server

IoT Server

None

Home Gateway

Remote Server

Server Address

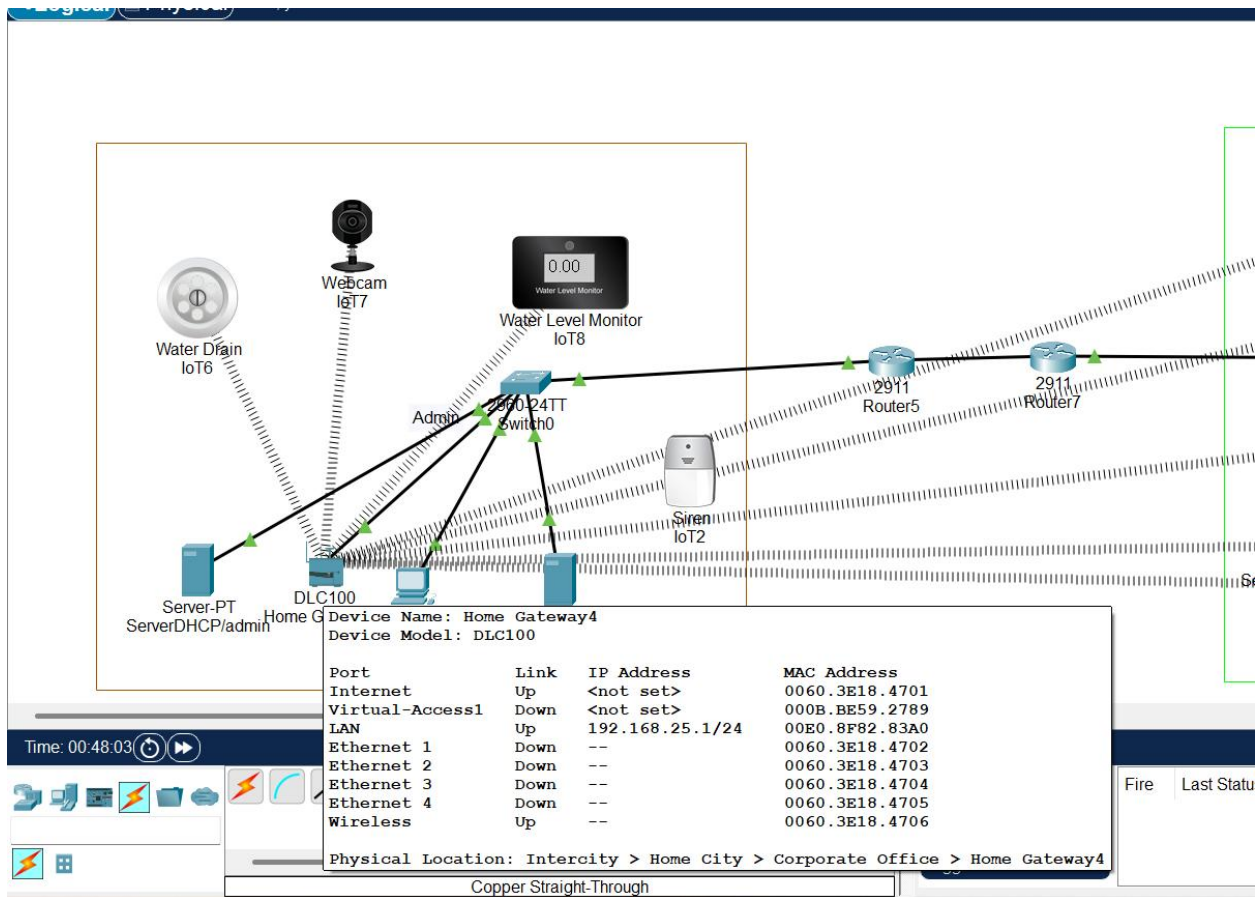
User Name

Password

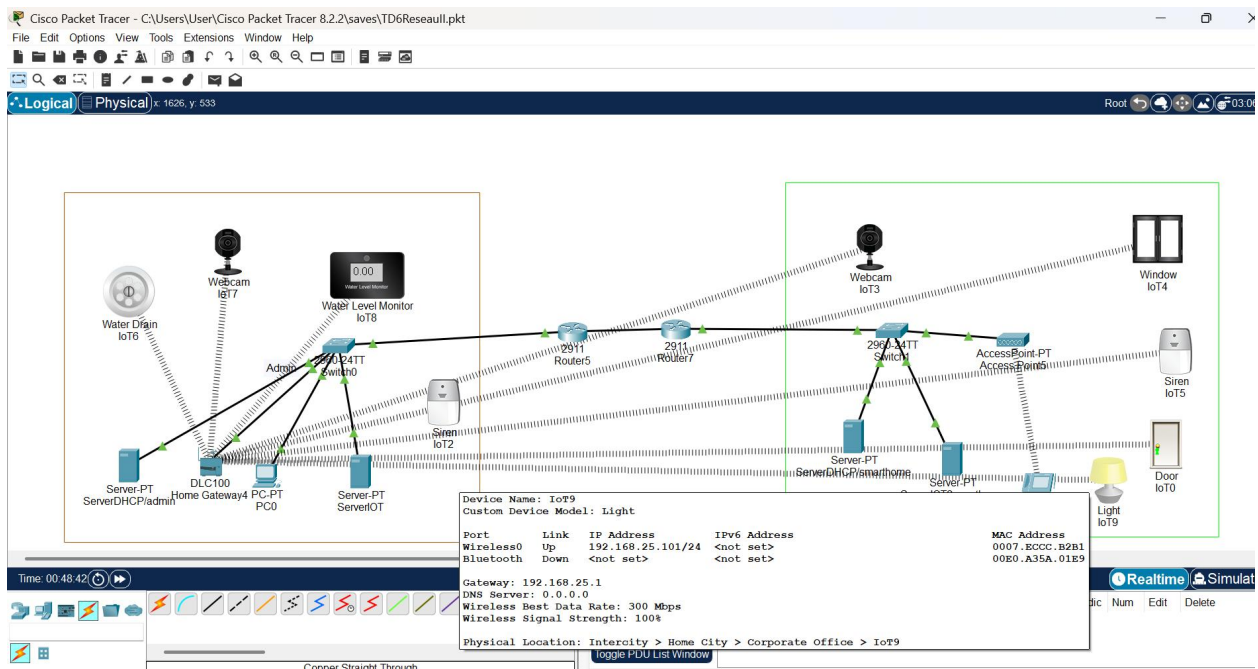
Connect

Top

Advanced



Le light prend en compte l'address ip du Home Gateway comme voulu



Sécurité du réseau IoT

Home Gateway3

Physical

Config

GUI

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Internet

LAN

Wireless

Wireless Settings

SSID

HomeGateway

2.4 GHz Channel

6 - 2.437GHz

Coverage Range (meters)

250,00

Authentication

☐ Disabled

☐ WEP

☐ WPA-PSK

☒ WPA2-PSK

☐ WPA

☐ WPA2

WEP Key

PSK Pass Phrase

IoT@2026

RADIUS Server Settings

IP Address

Shared Secret

Encryption Type

AES

☐ Top

Physical Config GUI Attributes**GLOBAL**

Settings

Algorithm Settings

INTERFACE

Internet

LAN

Wireless

Wireless Settings

SSID admin

2.4 GHz Channel 6 - 2.437GHz

Coverage Range (meters) 250,00

Authentication

☐ Disabled☐ WEP

WEP Key

☐ WPA-PSK☒ WPA2-PSK

PSK Pass Phrase

IoT@2026

☐ WPA☐ WPA2

RADIUS Server Settings

IP Address

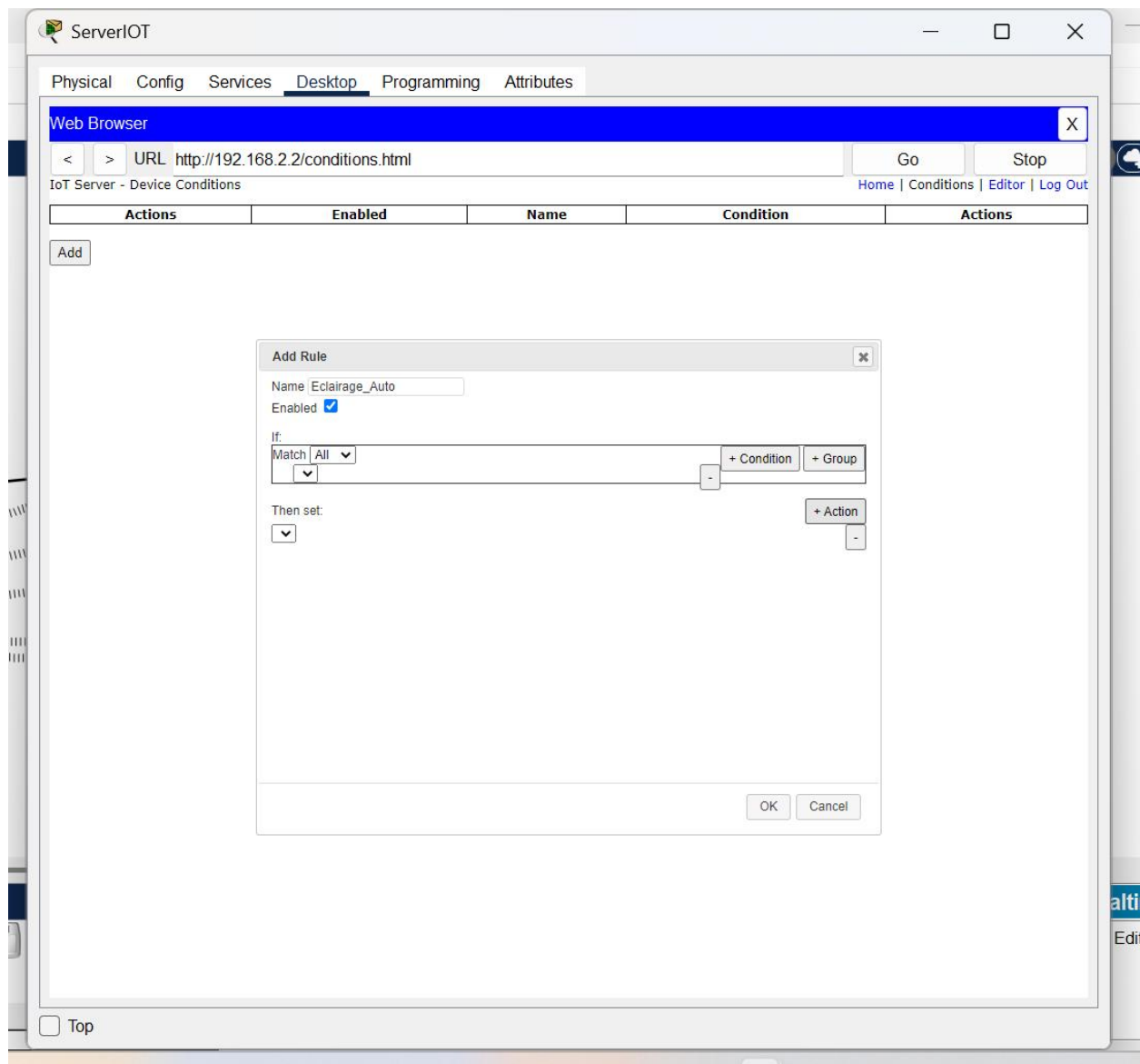
Shared Secret

Encryption Type

AES

☐ Top

2. Automatisation (IoT Rules)



Conclusion

Ce TD porte sur la mise en place d'une infrastructure de messagerie et d'un réseau domestique intelligent (**Smart Home**) pour l'**IUS Haïti.**, en simulant sur Cisco Packet Tracer un écosystème IoT sécurisé par **WPA2-PSK**. L'objectif était d'interconnecter des objets connectés à un serveur central (IP 192.168.2.2) pour automatiser des actions via des règles logiques.